

SYNTHETIC TEST PROPOSAL — NOT AN OFFICIAL CIL/NaCCER DOCUMENT

Project Title: AI-Based Real-Time Methane Monitoring System

Problem Statement

Methane gas buildup in underground coal seams poses severe explosion risks.

Project Objectives

1. Deploy spatial IoT gas sensors in underground coal mine galleries.
2. Predict methane ventilation thresholds using edge machine learning models.

Proposed Methodology

Hardware sensor node deployment integrated with spatial mesh networking.

Expected Outcomes

1. Real-time CH₄ alert system with sub-minute latency.
2. Reduced mine ventilation energy costs.

Estimated Cost

Total Budget: Rs 48.50 Lakhs

Equipment: Rs 20.00 Lakhs

Personnel: Rs 18.50 Lakhs

Consumables: Rs 10.00 Lakhs